Abdenour Foudhaili

Software Engineering Student | Seeking PFE Internship – Full Stack Developer (Angular / Spring Boot)
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Summary

3rd-year software engineering student at ESPRIT, seeking a **PFE internship** as a **Full Stack Developer**. Skills in **Java**, **Spring Boot**, **Angular** and **microservices** architectures, complemented by **NestJS**, REST APIs and **RabbitMQ**. Experience in **DevOps** (Docker, Jenkins, Kubernetes), automated testing (JUnit/Mockito) and development of code analysis and AI-based test generation tools.

Education

École Supérieure Privée d'Ingénierie et de Technologies (ESPRIT) — Ariana, Tunisia
Engineering Cycle in Computer Science – Software Engineering — *09/2024 – Present*

Institut Supérieur d'Informatique du Kef (ISIK) — Le Kef, Tunisia
Bachelor's Degree in Computer Science — *09/2019 – 08/2023*

Professional Experience

ESPRIT – Information Systems Department (DSI) — Ariana, Tunisia
Engineering Intern (PFA) – Scrum Master & Backend Developer — *06/2026 – 08/2026*
DigiFederation — Sports Management Platform for a Tunisian Sports Federation
Project being prepared for production release

- Developed the **Competitions module** with **NestJS** and **MongoDB**: registrations, participant eligibility, venue and referee management, schedules and rankings.
- Designed and configured the **API Gateway** in **Node.js** for request routing to microservices and CORS management; contributed to the Finance and License modules, with inter-service integration via **HTTP/REST** and **RabbitMQ**.
- Led a team of **6 members** as **Scrum Master**: Sprint Backlog tracking, blocker resolution and service integration coordination.

Bee Coders — Remote
DevOps Intern — *08/2025 – 09/2025*

- Containerized an **Angular/Spring Boot** application with **Docker** and set up a **CI/CD** pipeline with **Jenkins** to automate build and deployment.
- Deployed the application on **Kubernetes** (Pods/Services), handled monitoring with **Grafana** and analyzed code quality with **SonarQube**.

Projects

AI Code Reviewer – AI-Powered Code Analysis and Test Generation Platform | NestJS, Docker, Tree-sitter, Semgrep, PostgreSQL, pgvector
Automated GitHub Pull Request analysis platform combining code analysis and AI to detect issues and post contextual comments directly on code changes.

- Developed an automated Java test generation pipeline with **Tree-sitter**, **JUnit 5**, **Mockito** and **LLMs**, including secure compilation and execution in a **Docker sandbox**.
- Optimized code analysis with **Semgrep**, routing between **Gemini** and **Claude**, and a SHA-256-based **PostgreSQL** cache to reduce LLM calls and token consumption.
- Implemented asynchronous processing with **Redis/BullMQ**, **NestJS** REST APIs and automatic publication of reports on GitHub Pull Requests; set up automated tests with **Vitest** and E2E tests with **Katalon Recorder (Selenium)**.

E-Health Platform for Alzheimer's Risk Detection | Spring Boot, Angular, WebSocket, Keycloak, OpenFeign, RabbitMQ, Docker, Jenkins, Nexus, Kubernetes, SonarQube, Prometheus, Grafana
Microservices-based platform to monitor Alzheimer's patients and detect at-risk situations in real time.

- Developed the **Daily Assistance microservice**: REST API, medical reports, assisted memory and handover sheets.
- Integrated the microservice with other services via **OpenFeign** and **RabbitMQ**, and implemented real-time notifications with **WebSocket** as well as the **Angular** interfaces.
- Contributed to the **DevOps** pipeline: Docker, Jenkins CI/CD, Nexus, Kubernetes, SonarQube and Prometheus/Grafana monitoring; wrote tests with JUnit and Mockito.

FlashDrop – Flash Sales E-Commerce Platform | Spring Cloud, Angular, RabbitMQ, OpenFeign
Microservices-based e-commerce platform coordinating orders, stock and payments during flash sales.

- Developed the **Orders, Stock, Payments and Product Catalog** microservices.
- Developed the Flash Sale Management module and integrated inter-service communication with **RabbitMQ** and **OpenFeign** to coordinate order, stock and payment processes.

Choufli Hal – Corporate Plan and Incident Management Platform | Java, Symfony 6.4, JavaFX
Incident management platform for reporting, tracking and resolving incidents, with interfaces tailored to users, administrators and super administrators.

- Developed plan management features with Symfony and JavaFX.
- Integrated e-mail/SMS notifications via JavaMail and Twilio, as well as pagination and advanced sorting.

Technical Skills

Languages : Java, TypeScript, JavaScript, SQL, PHP
Frameworks : Spring Boot, Spring Cloud, NestJS, Node.js, Next.js, Angular, JPA/Hibernate, Symfony, JavaFX
Architecture & Systems : Microservices, API Gateway, REST API, WebSocket, RabbitMQ, OpenFeign, Keycloak
Databases : PostgreSQL, Redis, MySQL, MongoDB
DevOps & Quality : Docker, Kubernetes, Jenkins (CI/CD), Nexus, SonarQube, Semgrep (SAST), Prometheus, Grafana
Tools & Methods : Git, Maven, Vitest, Selenium (Katalon Recorder), JUnit, Mockito, Tree-sitter, Agile, Scrum, Power BI

Certifications

Bee Coders: Certificate of Achievement, DevOps training and coaching (2 months) – 10/2025
NVIDIA: Fundamentals of Deep Learning – 04/2025 (Credential ID : lir0qKNTQ-O4WDiCEhutUQ)
Dauphine | PSL Tunis – Digital Training × Elife (DTxE): Full Stack Java / Spring Boot – 01/2024 – 07/2024
Bee Coders: Angular / Spring Boot Training – 10/2023 – 12/2023

Languages

Arabic (native) • French (B2) • English (B2)